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**B.Sc. (E.C.S.) (Semester - II) (New) (NEP CBCS) Examination:
March/April – 2025
Introduction to Microcontroller and Embedded System (G06-0203)**

Day & Date: Saturday, 17-05-2025
Time: 12:00 PM To 01:30 PM

Max. Marks: 30

Instructions: 1) All questions are compulsory.
2) Figures to the right indicate full marks.

Q.1 Choose correct alternatives from the options. 06

- 1) 8051 Microcontroller is _____ Bit.
 - a) 8
 - b) 16
 - c) 32
 - d) 64
- 2) The internal RAM memory of the 8051 is _____.
 - a) 32 bytes
 - b) 64 bytes
 - c) 128 bytes
 - d) 256 bytes
- 3) How an embedded system communicate with the outside world?
 - a) Memory
 - b) Output
 - c) Peripherals
 - d) Input
- 4) When the microcontroller executes some arithmetic operations, then the flag bits of which register are affected?
 - a) PSW
 - b) SP
 - c) DPTR
 - d) PC
- 5) 8051 microcontrollers are manufactured by which of the following companies?
 - a) Atmel
 - b) Philips
 - c) Intel
 - d) All of the mentioned
- 6) Address bus of 8051 Microcontroller is _____ Bit.
 - a) 8
 - b) 16
 - c) 32
 - d) 64

Q.2 Answer the following. (Any Three) 06

- a) Define embedded system?
- b) Define Data transfer Instruction?
- c) Write note on SFR in 8051 Microcontroller.
- d) Write note on Logical Instruction in 8051 Microcontroller.
- e) Explain any 2 characteristics of embedded system?

- Q.3 Answer the following. (Any Two) 06**
- a) Explain difference between Microprocessor and Microcontroller?
 - b) Write note on Keil Micro vision Simulation tools.
 - c) Explain Clock and Reset circuit in 8051 Microcontroller?
- Q.4 Answer the following. (Any Two) 06**
- a) Write down features of 8051 Microcontroller.
 - b) Explain different types of Embedded System?
 - c) Write Embedded c program for LED blinking.
- Q.5 Answer the following. (Any One) 06**
- a) Draw and Explain Block diagram of 8051 Microcontroller?
 - b) Draw and Explain Architecture of Embedded System?